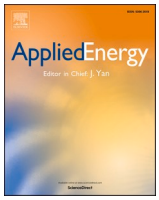
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# Collaborative optimization of renewable energy power systems integrating electrolytic aluminum load regulation and thermal power deep peak shaving

Xiaoyu Yue [a], Siyang Liao [a,*], Jian Xu [a], Deping Ke [a], Huiji Wang [a], Jiaquan Yang [b], Xuehao He [b]

[a] Hubei Engineering and Technology Research Center for AC/DC Intelligent Distribution Network, School of Electrical Engineering and Automation, Wuhan University, Wuhan 430072, China
[b] Electric Power Research Institute of Yunnan Power Grid Co., Ltd, Kunming 430072, China

## HIGHLIGHTS

- A refined peak regulation model of the electrolytic aluminum load (EAL) is established.
- The segmented cost model of the EAL is constructed.
- Collaborative optimization integrating EAL regulation and thermal power deep peak shaving.
- The effectiveness of collaborative optimization under various RE scenarios is proved.
- The influence of EAL regulation space on the social peak regulation effect is discussed.

## ARTICLE INFO

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## ABSTRACT

Addressing renewable energy (RE) curtailment in power systems necessitates a comprehensive strategy leveraging peak regulation resources from both the power and load sides. On the power side, deep peak shaving of thermal power plants can mitigate surplus electricity during periods of high RE production. On the load side, energy-intensive industrial loads, characterized by large capacity and rapid adjustability, can respond effectively to energy deficits when RE is low. To enhance the peak regulation capacity for optimal RE accommodation, this paper proposes a collaborative optimization method combining electrolytic aluminum load (EAL) regulation with thermal power deep peak shaving (DPS). The study is initiated by developing a sophisticated peak regulation model for the electrolytic aluminum load (EAL), considering production characteristics, cost features, and safety constraints. Subsequently, a collaborative optimization model is formulated, integrating EAL regulation with thermal power deep peak shaving (DPS), aiming to minimize societal peak regulation costs (PRC). Applying this model to a practical regional grid in Yunnan Province reveals substantial reductions in RE curtailment rates, effective minimization of total social PRC, and enhanced operational economics of thermal power DPR under varying wind power scenarios. Notably, the proposed approach requires only minor adjustments to the EAL, ensuring production safety is maintained. This research provides valuable insights for the seamless integration of EAL into grid peak regulation services.

## 1. Introduction

Nowadays, all countries in the world are working hard to cope with the challenges of fossil energy shortage and excessive carbon emissions [1–3]. It has become a global consensus to develop clean and low-carbon renewable energy sources such as wind energy and solar energy [4]. However, the inherent randomness, volatility, and intermittency of wind and solar power pose significant challenges to RE accommodation [5]. Notably, wind power's characteristic counter-peak behavior aggravates fluctuations in the net load and peak-to-valley differences within power systems, substantially increasing system operation costs and the burden on peak regulation services [6–9].

* Corresponding author.
*E-mail address:* liaosiyang@whu.edu.cn (S. Liao).

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0306-2619/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

To exploit peak regulation resources from both power and load sides has garnered significant attention from scholars worldwide. On the power side, thermal power plants serve as the primary source for peak regulation in China, with deep peak shaving (DPS) becoming standard practice. Following flexibility renovations, the minimum stable combustion load for thermal power units in China reaches 30% to 35%, while in some advanced units, it can be as low as 20% to 25% [10–12]. In regional power grids with a relatively low penetration of RE, DPS from thermal power plants can effectively support to accommodate RE. However, as the proportion of renewable energy sources connected to the grid increases, relying solely on DPS from thermal power is no longer enough. In certain regions, significant wind and solar curtailment have emerged, while the economic performance of thermal power plants has also been significantly undermined. In particular, in the Yunnan region of China, where thermal power capacity accounts for a relatively low proportion, the peak regulation capability is severely inadequate in the face of the yearly rising trend of RE installations. Simultaneously, guided by national policies to reduce carbon emissions and energy consumption per unit of GDP, there has been a large relocation of energy-intensive industrial loads (EIL) to central and western provinces in recent years. In the Yunnan region, for instance, a large volume of these EIL has been absorbed, with the proportion of such industrial loads in many regional power grids significantly exceeding that of conventional loads (CL). On the load side, compared to conventional flexible loads such as residential and commercial building loads, EIL-like electrolytic aluminum (EA) displays large capacity and rapid adjustability characteristics, making them own great peak regulation potential. To lessen system peak regulation pressure, the Energy Bureau of Yunnan Province has introduced a scheme to organize grid peak regulation drills involving 16 EA users across the province. Besides, EIL can provide substantial flexibility to the system with only slight adjustments, serving as a strategic lever in critical. From here, developing the refined peak regulation models for EIL, considering their production adjustment characteristics and safety constraints, is of great significance for power systems with a high penetration of RE sources. This research also promises large socioeconomic benefits.

Current research on EIL mainly focuses on two aspects: 1) To schedule the power consumption for such industries to minimize electricity costs and ensure business profitability. For instance, research has been conducted to evaluate the potential of external power scheduling through optimizing steel plant production processes [13]. Studies have also focused on tracking electricity consumption in steel mills [14,15], deploying self-generation equipment [16], and implementing flexible resource scheduling within resource-task networks [17] to minimize imbalance costs and overall energy expenses. Additionally, considering the time constraints for production processes in steel plants, multi-objective optimization models are developed in systems integrating RE sources and energy storage systems to balance costs, emissions, and production schedules [18]. Moreover, investigations have looked into privacy protection and robust aggregation of multiple EIL entities participating in grid dispatching [19,20], as well as their strategic engagement in day-ahead spot markets and reserve markets [21,22] to maximize economic benefits. 2) This research explores the control methodologies for EIL to effectively partake in grid frequency regulation, thereby stabilizing wind power fluctuations. Typically, feedback control mechanisms are designed considering parameters such as voltage, current, and frequency to regulate the power of EALs in response to wind power variability [23–25]. A majority of studies focus on electrolysis aluminum load (EAL), employing methods like direct adjustment of their saturated reactor parameters [26], designing a combined regulation scheme for saturated reactors and transformer tap changers [27], or coordinating control with generator excitation systems [28]. Overall, there is a scarcity of research on the optimization of EILs participating in grid peak regulation operations. Additionally, the model of EILs has typically been oversimplified, employing interrupted and adjustable load models that inadequately capture the actual production and regulation characteristics of these loads, and the cost features are also not thoroughly investigated. While some studies have developed detailed production and regulation characteristic models for electric arc furnace magnesium loads [29], it should be noted that there are significant differences in the production properties between arc furnace-type and electrolysis-type loads, rendering the regulation models non-universal. This study specifically focuses on EAL.

Furthermore, EAL regulation and thermal power DPS exhibit a certain degree of complementary nature. Thermal power plants can effectively address surplus electricity during periods of high RE production through DPS, while EILs can reduce their consumption when RE is low or modestly increase it when there's an excess supply. The synergistic optimization of these two mechanisms can significantly enhance the bidirectional peak regulation capacity for the grid. However, there is currently a dearth of research that comprehensively considers the coordination and optimization of both the regulation potential of EAL and the DPS capabilities of thermal power units, particularly in regional grids with high penetrations of RE sources and EAL. Therefore, this study proposes an innovative refined adjustment model for EAL participation in peak regulation of power grid, considering the complex production process and multi-factor coupling constraints. It investigates the segmented regulation cost characteristics of EAL and introduces a coordinated optimization approach integrating EAL with thermal power DPS, which enhances the peak shaving effect of the power grid and facilitates a higher level of RE accommodation.

In summary, this paper first develops a refined peak regulation model for EAL. Second, it proposes a collaborative optimization model combining the regulation of EAL with thermal power DPS, aiming to minimize the overall societal peaking costs. Afterward, using a regional power grid in Yunnan Province with a high proportion of renewable energy sources as an example, the paper verifies the comprehensive benefits of the proposed model. Fig. 1 shows the framework and contributions of this study. The main contributions are:

(1) Considering the detailed production characteristics, controllability, comprehensive cost features, and safety constraints of EAL, this study develops a refined peak regulation model that integrates power, current, temperature, and production efficiency. This model ensures safe and reliable load regulation for EAL.
(2) A collaborative optimization model is established that combines the regulation of EAL with thermal power DPS. The model determines the peak regulation status of individual EAL and thermal power units, taking into account the comprehensive costs associated with EAL adjustments, the economic viability of thermal power DPS, and penalties for RE curtailment, aiming to balance overall societal economic benefits with RE accommodation.
(3) Taking a real-world power grid as an example, the paper verifies the effectiveness of the proposed model under three typical wind power scenarios. Analyzing the great comprehensive benefits that adjusting EAL can bring to the entire societal power systems, with only minor adjustments needed within the bounds of their production characteristics and safety constraints. Exploring the impact of different levels of adjustability in EAL on the overall benefits.

## 2. Modeling of electrolytic aluminum load

### 2.1. Modeling of regulation characteristics of EAL

The production principle of the modern EA industry is to put hundreds of kilowatts of direct current into the electrolytic cell, produce high-temperature environment to smelt molten alumina, and produce metal aluminum by chemical reaction [30]. Maintaining a precise heat balance is crucial in EA production, requiring the electrolytic cell to consistently operate within a temperature range of 950 °C to 970 °C, regulated by the control of direct current. The electrolytic cell possesses

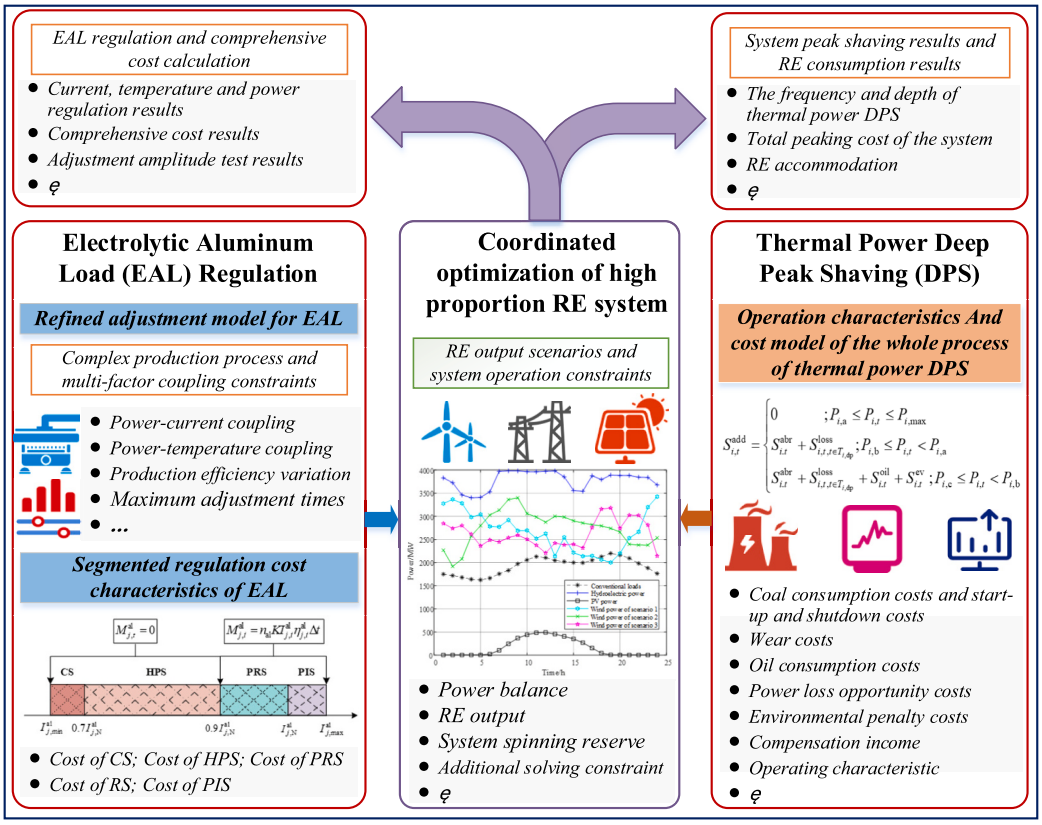



**Fig. 1.** The framework and contributions of this study.

significant thermal inertia, with a heat time constant reaching several hours [31]. The most serious situation that the electrolytic cell needs to prevent is that a long-term power outage causes the cryolite to solidify in the electrolytic cell. However, even short-term outages lasting from a few minutes to three hours do not pose a risk of cryolite solidification. Thus, power fluctuations within specified temporal bounds and magnitude thresholds do not compromise the secure and uninterrupted production processes within electrolytic cells.

In the context of actual industrial processes, the electricity demand of EA production is effectively managed and adjusted through two key strategies: 1) The EAL is capable of coarse regulation through manipulation of the tap positions on an on-load tap-changing transformer. A single tap change results in a 3% alteration in nominal power, and such transitions occur within a swift 7-s interval. Consequently, it is possible to effectuate changes amounting to nearly 30% of the system's rated capacity within approximately 70 s. 2) The EAL also permits fine-grained control via the manipulation of a controllable saturated reactor. Post-modification, this reactor facilitates continuous variation across tap settings with an approximate 10% load regulation capability relative to its nominal power rating. It has been demonstrated that a 5.5% change in the load can be achieved within 2 s, while a full 10% adjustment is attainable within a 5-s interval [30]. Currently, EAL facilities in regions like Yunnan, Shandong, Qinghai, Guangxi, and others in China exhibit an adjustment capability of 5–10% upward and 30% downward at the rated power. This adjustment speed is robust enough to align with the requirements of hourly scheduling plans. In addition, the EA production series is operated by hundreds of electrolytic cell units in series mode. If you completely start and stop a production series, it will take several months, and the costs of re-opening the slot are as high as millions. Usually, the electrolytic aluminum plant does not stop, 8760 h a year to keep running.

In summary, the EAL is regarded as a load that can be continuously adjusted but cannot be interrupted within a certain range. The detailed modeling is carried out below.

### 2.1.1. Power-current coupling constraints

For EA plants, there are inherent upper and lower limits of EAL power. However, in practice, the EA plant pays more attention to the size of the series current, and the current is limited to a range. According to the common equivalent circuit diagram of EAL, there is an approximate square relationship between power and current at each moment. The power-current coupling constraint is as follows:

$$P^{\mathrm{al}}_{\min} \le P^{\mathrm{al}}_{j,t} \le P^{\mathrm{al}}_{\max} \tag{1}$$

$$I^{\mathrm{al}}_{\min} \le I^{\mathrm{al}}_{j,t} \le I^{\mathrm{al}}_{\max} \tag{2}$$

$$P^{\mathrm{al}}_{j,t} = I^{\mathrm{al}2}_{j,t} R_{\mathrm{m}} + I^{\mathrm{al}}_{j,t} E_{\mathrm{m}} \tag{3}$$

where $I^{\mathrm{al}}_{j,t}$ and $P^{\mathrm{al}}_{j,t}$ are the current and power of electrolytic aluminum series respectively. Considering the day-ahead scheduling optimization, $t$ is taken as the hour level. $R_{\mathrm{m}}$ and $E_{\mathrm{m}}$ are equivalent resistance and equivalent back electromotive force respectively, which are generally determined by specific EA plant parameters.

### 2.1.2. Power-temperature coupling constraints

A certain range of power changes in a certain period of time will not damage the heat balance of the electrolytic cell. The heat balance can be described by the following equation:

$$W_{\mathrm{in}} = \Delta H_0 + Q_{\mathrm{Out}} \tag{4}$$

$$W_{\mathrm{in}} = \int P^{\mathrm{al}}_{j,t} dt \tag{5}$$

where $W_{\mathrm{in}}$ is the heat input in the electrolytic cell, which is provided by the input power; $\Delta H_0$ is the heat consumed in the production process;

$Q_{\mathrm{Out}}$ is the heat lost in the production process.

When the power of the EAL changes, the heat balance formula changes as follows:

$$c_{\mathrm{al}} m_{\mathrm{al}} \Delta T_{\mathrm{al}} = \left(p_{j,t}^{\mathrm{al}} + \Delta P_{j,t}^{\mathrm{al}}\right) t - \Delta H_0 - Q_{\mathrm{Out}} \tag{6}$$

where: $c_{\mathrm{al}}$ and $m_{\mathrm{al}}$ are the specific heat capacity coefficient and mass of the electrolyte, respectively; the temperature change of the electrolytic cell is:

$$\Delta T_{\mathrm{al}} = \frac{\Delta P_{j,t}^{\mathrm{al}}}{c_{\mathrm{al}} m_{\mathrm{al}}} \tag{7}$$

Then the power-temperature coupling constraints for adjacent moments are as follows:

$$\left(P_{j,t}^{\mathrm{al}} - P_{j,t-1}^{\mathrm{al}}\right) \Delta t = c_{\mathrm{al}} m_{\mathrm{al}} \left(T_{j,t}^{\mathrm{al}} - T_{j,t-1}^{\mathrm{al}}\right) \tag{8}$$

$$T_{\min}^{\mathrm{al}} \le T_{j,t}^{\mathrm{al}} \le T_{\max}^{\mathrm{al}} \tag{9}$$

At any time, the upper and lower limits of the EA power adjustment generated by the power-temperature coupling relationship are:

$$\begin{aligned} P_{j,t,min}^{\mathrm{al-tem}} &= P_{j,t-1}^{\mathrm{al}} + c_{\mathrm{al}} m_{\mathrm{al}} \left(T_{\min}^{\mathrm{al}} - T_{j,t-1}\right) \Big/ \Delta t \\ P_{j,t,max}^{\mathrm{al-tem}} &= P_{j,t-1}^{\mathrm{al}} + c_{\mathrm{al}} m_{\mathrm{al}} \left(T_{\max}^{\mathrm{al}} - T_{j,t-1}\right) \Big/ \Delta t \end{aligned} \tag{10}$$

#### 2.1.3. State uniqueness constraint

For the production regulation of EA, there are three states: power holding state, power up-regulation state and power down-regulation state. Three molar variables $\lambda_{j,t}^{\mathrm{keep}}$, $\lambda_{j,t}^{\mathrm{up}}$, $\lambda_{j,t}^{\mathrm{down}}$ are introduced to indicate whether the EAL is in the above three states, taking 1 as yes and 0 as no. Then there exists a state uniqueness constraint at time $t$.

$$\lambda_{j,t}^{\mathrm{keep}} + \lambda_{j,t}^{\mathrm{up}} + \lambda_{j,t}^{\mathrm{down}} = 1 \tag{11}$$

#### 2.1.4. Power variation and climbing constraints

When the EAL is in the power holding state, that is, the load power at the adjacent time does not change, then $\lambda_{j,t}^{\mathrm{keep}} = 1$; conversely, when the load is in a regulated state, then $\lambda_{j,t}^{\mathrm{keep}} = 0$. The power variation and climbing constraints at adjacent moments are as follows:

$$P_{j,t}^{\mathrm{al}} = P_{j,t-1}^{\mathrm{al}} + \Delta P_{j,t}^{\mathrm{al}} \tag{12}$$

$$\lambda_{j,t}^{\mathrm{up}} R_{\mathrm{down}}^{\mathrm{al}} \le \Delta P_{i,t}^{\mathrm{al}} \le \lambda_{j,t}^{\mathrm{up}} R_{\mathrm{up}}^{\mathrm{al}} \tag{13}$$

where $R_{\mathrm{down}}^{\mathrm{al}}$ and $R_{\mathrm{up}}^{\mathrm{al}}$ are the load regulation speed of electrolytic aluminum in each time step. For the hour-level scheduling period, the adjustment speed is sufficient to support the large-scale power adjustment according to the practical test results. Considering the safety conservatism of the practical application, the hour-level adjustment speed is set to 20% of the rated power.

#### 2.1.5. Maximum adjustment times constraint

In order to prevent the mechanical parts of the equipment from being worn due to too many times of EAL adjustment, the maximum adjustment times in a scheduling cycle should be limited.

$$\sum_{t=1}^{T} \left(\lambda_{j,t}^{\mathrm{up}} + \lambda_{j,t}^{\mathrm{down}}\right) \le N_j^{\mathrm{al\text{-}rel}} \tag{14}$$

where: $N_j^{\mathrm{al\text{-}rel}}$ is the maximum allowable adjustment times of EAL in a scheduling cycle $T$.

#### 2.1.6. Continuity adjustment constraints

Due to the fact that continuous adjustments of EAL can lead to frequent fluctuations in electrolytic cell temperatures, and considering practical production processes, it is essential to maintain maximum operational stability. Hence, continuous adjustment at two adjacent moments are not allowed, including continuous upward adjustment, continuous downward adjustment, both upward adjustment and downward adjustment. Moreover, the load power after each adjustment needs to be kept for a minimum period of time $T_j^{\mathrm{keep}}$. Then there are the following constraints:

$$\left(\lambda_{j,t}^{\mathrm{up}} + \lambda_{j,t}^{\mathrm{down}}\right) + \left(\lambda_{j,t-1}^{\mathrm{up}} + \lambda_{j,t-1}^{\mathrm{down}}\right) \le 1 \tag{15}$$

$$\left(\lambda_{j,t-1}^{\mathrm{keep}} - \lambda_{j,t}^{\mathrm{keep}}\right)\left(T_{j,t-1}^{\mathrm{keep}} - T_j^{\mathrm{keep}}\right) = 1 \tag{16}$$

#### 2.1.7. Production efficiency variation constraint

Under normal operating conditions, EAL do not participate in grid scheduling optimizations and run at a constant power mode with a rated current efficiency of approximately 94%. However, when these loads are involved in system peak regulation optimization, power adjustments lead to temperature variations which directly affect the current efficiency of the electrolytic cell. Relevant studies have confirmed that for every decrease of 10 °C in the electrolytic cell temperature, the current efficiency improves by 1% ~ 1.5% [32,33]. Therefore, the following formula can be derived:

$$\eta_{j,t}^{\mathrm{al}} = \eta_{j,\mathrm{N}}^{\mathrm{al}} - 1.5\%\left(T_{j,t} - T_{j,\mathrm{N}}\right) \tag{17}$$

where $\eta_{j,\mathrm{N}}^{\mathrm{al}}$ is the rated current efficiency of EAL; $T_{j,\mathrm{N}}$ is the rated temperature of the electrolytic cell.

#### 2.1.8. Production output state determination constraints

The series current $I_{j,t}^{\mathrm{al}}$ of the electrolytic aluminum load is the key parameter of electrolytic process, which directly affects the output of electrolytic aluminum. Since the electrolytic aluminum load is composed of hundreds of electrolytic cells in series, the output is the sum of the output of the electrolytic cells in series, and the output per unit time period is calculated as follows:

$$M_j^{\mathrm{al}} = n_{\mathrm{al}} K_{\mathrm{al}} I_{j,t}^{\mathrm{al}} \eta_{j,t}^{\mathrm{al}} \Delta t \tag{18}$$

where $n_{\mathrm{al}}$ is the number of electrolytic cells; $K_{\mathrm{al}}$ is the electrochemical equivalent of EA, the unit is $g/(\mathrm{A{\cdot}h})$, which can be taken as 0.3356. Moreover, the output is generally positively correlated with the current size.

Usually, the series current is kept near the rated value to ensure the safety and stability of production. However, when the EAL is reduced, the current decreases, which in turn affects the production yield. Only when the series current of electrolytic aluminum load is greater than 90% of the rated current, there will be aluminum production. There are the following constraints:

$$M_{j,t}^{\mathrm{al}} = \begin{cases} 0, & if \quad I_{j,t}^{\mathrm{al}} \le 0.9 I_{j,\mathrm{N}}^{\mathrm{al}} \\ n_{\mathrm{al}} K_{\mathrm{al}} I_{j,t}^{\mathrm{al}} \eta_{j,t}^{\mathrm{al}} \Delta t, & if \quad I_{j,t}^{\mathrm{al}} > 0.9 I_{j,\mathrm{N}}^{\mathrm{al}} \end{cases} \tag{19}$$

#### 2.1.9. Yield constraint

To ensure the economic benefits of the EA plant to the greatest extent possible, the order demand within the scheduling period must be guaranteed. In theory, there should exist an acceptable margin where production is slightly less or slightly more than the target. The formula is expressed as follows:

$$\beta_{\min} M_{j,\mathrm{N}}^{\mathrm{al}} \le \sum_{t}^{T} M_{j,t}^{\mathrm{al}} \le \beta_{\max} M_{j,\mathrm{N}}^{\mathrm{al}} \tag{20}$$

where $M_{j,\mathrm{N}}^{\mathrm{al}}$ is the rated output of electrolytic aluminum load at a single

time interval; $\beta_{\min}$ and $\beta_{\max}$ are the minimum and maximum coefficients of the load yield range, which are 0.98 and 1.02, respectively.

#### 2.1.10. Cooling state maximum duration constraint

When the series current falls to 70% or less of its rated value, the EAL enters a cooling state, which results in a decrease in the electrolysis cell temperature. Consequently, the molten electrolyte gradually cools down, potentially posing risks to equipment safety. Based on operational data from an actual EA plant, during a total power outage, it takes approximately 7 to 8 h in summer and 3 to 4 h in winter for the electrolysis cells to fully solidify. Therefore, under exceptional circumstances where the EAL is inadvertently shifted into a cooling state, it is crucial to strictly limit the duration of this cooling period to prevent cell solidification. Here the state variable $y_1$ is introduced to indicate whether the load is in the cooling state, taking 1 as yes and 0 as no. The maximum time $T_{\text{cool}}$ of cooling state is constrained as:

$$\sum_{a=1}^{T-T_{\text{cool}}+1} \prod_{t=a}^{a+T_{\text{cool}}} y_{1,t} = 0 \tag{21}$$

### 2.2. Modeling of the costs characteristic of EAL

Considering the comprehensive power regulation process for EAL, employing series current intensity as the criterion, the load's operational states are delineated into five distinct stages: Cooling State (CS), Heat Preservation State (HPS), Production Reduction State (PRS), Rated Operation State (RS), and Production Increase State (PIS). The corresponding breakpoints for these states in terms of series current intensity are $I^{\text{al}}_{j,min}$, $0.7I^{\text{al}}_{j,\text{N}}$, $0.9I^{\text{al}}_{j,\text{N}}$, $I^{\text{al}}_{j,\text{N}}$ and $I^{\text{al}}_{j,max}$, respectively. Notably, the first three states transpire during load reduction. The visual representation of these state segments is depicted in Fig. 2 below. Significantly, aluminum production occurs exclusively when the EAL is adjusted within the specified green and purple ranges. And the power upward space is relatively small.

#### 2.2.1. Costs of CS

In this state where $I^{\text{al}}_{j,t} \le 0.7I^{\text{al}}_{j,\text{N}}$, the EAL still consumes power but produces no output. Therefore, there are electricity costs and production loss costs, calculated as follows:

$$S^{\text{cost}}_{j,t} = c_{\text{grid}} P^{\text{al}}_{j,t} \Delta t \tag{22}$$

$$S^{\text{loss}}_{j,t} = c_{\text{al}} M^{\text{al}}_{j,\text{N}} \tag{23}$$

where $c_{\text{grid}}$ is the time-of-use electricity price of the EAL purchased from the power grid; $c_{\text{al}}$ is the unit price of the spot market of the current season aluminum, which is $/ton.

In addition, due to the CS for a long time will lead to the solidification of the electrolytic cell, harming the electrolytic cell equipment, resulting in re-slotting, and bringing serious economic losses. This part of the costs belongs to the risk costs of the equipment condensing tank, and there is no clear calculation method at present. In theory, the risk function should be a convex function form of nonlinear change, which depends on the analysis and fitting of a large number of operating data

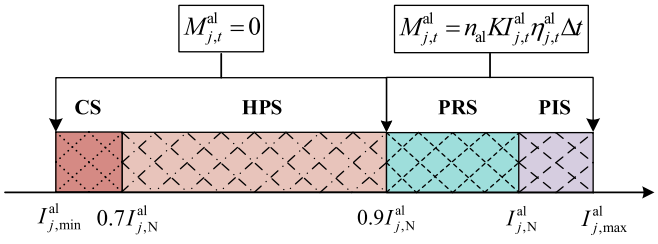



**Fig. 2.** Sectional diagram of EAL current intensity and load state.

in the cooling state, but there is currently a lack of risk data support. Finally, through the investigation of the metallurgical and material industries, this study refers to [34,35] to construct the risk function. The risk cost $S^{\text{wtr-cool}}_{j,t}$ at the time $t$ is calculated as follows:

$$S^{\text{wtr-cool}}_{j,t} = v^{\text{cool}}_{j,t} S^{\text{al}}_{j,\text{on}} \tag{24}$$

$$v^{\text{cool}}_{j,t} = 1 - \mathrm{e}^{-\gamma T^{\text{al}}_{j,t}}, \quad T^{\text{al}}_{j,t} \le T_{\text{cool}} \tag{25}$$

where $v^{\text{cool}}_{\text{al},j}$ is the risk probability of condensate tank event; $S^{\text{al}}_{j,\text{on}}$ is the complete start-stop cost of re-slotting equipment; $T_{\text{cool}}$ is the maximum non-condensing tank operation time of the equipment under cooling state, which can be set to 3 h; $T^{\text{al}}_{j,t}$ is the total duration of the EAL in the cooling state at the time $t$. Since entering the cooling state at any time before the time $t$ will cause cumulative damage to the equipment, the cumulative risk at the time $t$ in the scheduling cycle is included. $\gamma$ is the actual tank risk of EAL reaching the maximum running time in the cooling state, which can be taken as 95% according to production experience.

#### 2.2.2. Costs of HPS

When the series current drops below 90% of its rated value, the EAL enters HPS. During this phase, the electrolysis cell continues to consume power, solely aiming to sustain the molten electrolyte at an optimal temperature without contributing to any production output. Consequently, in this state, the incurred costs encompass only electricity costs and production loss costs. The calculation method aligns with the approach outlined in Eqs. (22) and (23).

#### 2.2.3. Costs of PRS

As the series current declines to 90%–100% of its rated value, the EAL transitions into a reduced production state. During this phase, aluminum production persists but at a proportionately diminished output rate. The associated costs encompass both electricity costs and production losses. Electricity costs are computed akin to Eq. (22), while estimating production loss costs involves the following:

$$S^{\text{loss-drop}}_{j,t} = c_{\text{al}}\left(M^{\text{al}}_{j,\text{N}} - M^{\text{al}}_{j,t}\right) \tag{26}$$

#### 2.2.4. Costs of RS

In the rated state, $I^{\text{al}}_{j,t} = I^{\text{al}}_{j,\text{N}}$, the EA power and output are rated values, only the electricity costs calculation is the same as the above Eq. (22).

#### 2.2.5. Costs of PIS

Numerous EA plants in China have conducted experiments to augment series current, aiming to boost production. When the series current surpasses the rated value, it escalates both EA and aluminum output, yielding increased production benefits. However, adherence to a two-part tariff system in EA enterprises introduces complexities. The capacity price, linked to monthly maximum load power, escalates if the load exceeds the initial value on any day. Consequently, augmenting EAL raises capacity price costs. Simultaneously, heightened current and power levels elevate the electrolytic cell temperature, risking accelerated wear and corrosion of equipment. This study mitigates safety risks by rigorously maintaining the electrolytic cell temperature within the range of 950 °C to 970 °C, preventing high-temperature incidents and minimizing equipment losses. Consequently, the accrued production benefits and additional capacity tariff costs are calculated as outlined below:

$$G^{\text{benefit}}_{j,t} = c_{\text{al}}\left(M^{\text{al}}_{j,t} - M^{\text{al}}_{j,\text{N}}\right) \tag{27}$$

$$S^{\text{incr-cap}}_{j} = C^{\text{v-cap}}_{j}\left(max\left\{P^{\text{al}}_{j,t}\right\} - P^{\text{al}}_{j,N}\right) \tag{28}$$

where $C_j^{\text{v-cap}}$ is the electricity price standard per unit capacity increase.

In summary, corresponding to the five power states and cost functions of EAL, the segmented costs model is constructed, and the costs of RS and PRS can be combined, the formula is as follows:

$$S_{j,t}^{\text{al}} = \begin{cases} S_{j,t}^{\text{cost}} + S_{j,t}^{\text{loss}} + S_{j,t}^{\text{wtr-cool}}, & I_{j,min}^{\text{al}} \leqslant I_{j,t}^{\text{al}} \leqslant 0.7 I_{j,\text{N}}^{\text{al}} \\ S_{j,t}^{\text{cost}} + S_{j,t}^{\text{loss}}, & 0.7 I_{j,\text{N}}^{\text{al}} < I_{j,t}^{\text{al}} \leqslant 0.9 I_{j,\text{N}}^{\text{al}} \\ S_{j,t}^{\text{cost}} + S_{j,t}^{\text{loss-drop}}, & 0.9 I_{j,\text{N}}^{\text{al}} < I_{j,t}^{\text{al}} \leqslant I_{j,\text{N}}^{\text{al}} \\ S_{j,t}^{\text{cost}} - G_{j,t}^{\text{benefit}} + S_j^{\text{incr-cap}}, & I_{j,\text{N}}^{\text{al}} < I_{j,t}^{\text{al}} \leqslant I_{j,max}^{\text{al}} \end{cases} \tag{29}$$

## 3. Collaborative optimization models and methods

### 3.1. Modeling of the whole process operation costs of thermal power DPS

#### 3.1.1. Coal consumption costs and start-up and shutdown costs

When thermal units are operated in the basic peaking range, typically 50% of rated capacity and above, there are only coal consumption costs and startup and shutdown costs, which are calculated as follows:

$$S_{i,t}^{\text{coal}} = a_i {P_{i,t}}^2 + b_i P_{i,t} + c_i \tag{30}$$

$$S_{i,t}^{\text{T}} = U_{i,t}\left(1 - U_{i,t-1}\right) C_{i,\text{on}} + U_{i,t-1}\left(1 - U_{i,t}\right) C_{i,\text{off}} \tag{31}$$

where $i$ is the serial number of the thermal power unit, $a_i$, $b_i$, $c_i$ is the characteristic coefficient of coal consumption after economic conversion; $P_{i,t}$ is the power of thermal power unit. $U_{i,t}$ characterizes the operating status of the unit, taking 1 as start-up and 0 as shutdown; $C_{i,\text{on}}$ and $C_{i,\text{off}}$ are the start-up and shutdown costs of the unit, respectively.

#### 3.1.2. Wear costs of DPS

When the thermal power unit is in the DPS stage, additional unit wear costs will be generated due to low-cycle fatigue loss and creep loss, which can be calculated as follows.

$$S_{i,t}^{\text{abr}} = \frac{1}{2N_{i,t}\left(P_{i,t}\right)} \tau S_{\text{unit}} \tag{32}$$

where $N_{i,t}$ is the rotor fracture cycle, the value is related to $P_{i,t}$, which can be calculated by the Manson-Coffin formula [12]; $S_{\text{unit}}$ is the purchase costs of the unit; $\tau$ is the actual operating loss coefficient of the unit; when considering the two-phase deep peaking of the unit, $\tau_1$ and $\tau_2$ are the wear coefficients of the first and the second phases, respectively.

#### 3.1.3. Oil consumption costs of DPS

When the thermal power unit is in the second stage of DPS, the combustion stability of the boiler decreases, and the unit needs to be fueled to ensure safe operation. The fuel consumption costs are calculated as follows:

$$S_{i,t}^{\text{oil}} = c_{\text{oil}} H_i^{\text{oil}} \tag{33}$$

where $H_i^{\text{oil}}$ is the fuel consumption of the unit; $c_{\text{oil}}$ is the fuel price in the current season.

#### 3.1.4. Power loss opportunity costs of DPS

When the thermal power unit engages in DPS, the opportunity costs of the lost electricity generation revenue should be included as part of the overall costs structure for the thermal power plant. The calculation is as follows:

$$S_{i,t,t\in T_{i,\text{dp}}}^{\text{loss}} = \left(\left(P_{i,\text{a}} - P_{i,t}\right)\Delta T\right) C_{\text{price,GC}} \tag{34}$$

where $C_{\text{price,GC}}$ represents the grid connection tariff.

#### 3.1.5. Environmental penalty costs of DPS

In China, with the increasingly stringent control of air pollutant emissions, it is explicitly mandated that coal-fired power plants undergoing flexibility renovations must be equipped with highly efficient desulfurization and denitrification facilities to ensure that their $SO_2$ and $NO_X$ emissions comply with the 'Emission Standards for Air Pollutants from Thermal Power Plants'. However, during the DPS period, the efficiency of environmental protection equipment may decrease, leading to excessive emissions of pollutants and resulting in penalties. The environmental penalty costs are:

$$S_{i,t}^{\text{ev}} = C_{\text{S}} V_{\text{S}} \Delta\mu_{t,i}^{\text{S}} + C_N V_{\text{N}} \Delta\mu_{t,i}^{\text{N}} \tag{35}$$

where $C_{\text{S}}$ and $C_N$ are the fines for $SO_2$ and $NO_X$ exceeding the standard in unit volume gas respectively; $V_{\text{S}}$ and $V_{\text{N}}$ are the emission standards of $SO_2$ and $NO_X$ in unit volume gas, respectively; $\Delta\mu_{t,i}^{\text{S}}$ and $\Delta\mu_{t,i}^{\text{N}}$ are the over-standard rates of $SO_2$ and $NO_X$ emissions, respectively.

Therefore, the total additional costs for DPS of thermal power units are:

$$S_{i,t}^{\text{add}} = \begin{cases} 0, & P_{i,\text{a}} \leqslant P_{i,t} \leqslant P_{i,max} \\ S_{i,t}^{\text{abr}} + S_{i,t,t\in T_{i,\text{dp}}}^{\text{loss}}, & P_{i,\text{b}} \leqslant P_{i,t} < P_{i,\text{a}} \\ S_{i,t}^{\text{abr}} + S_{i,t,t\in T_{i,\text{dp}}}^{\text{loss}} + S_{i,t}^{\text{oil}} + S_{i,t}^{\text{ev}}, & P_{i,\text{c}} \leqslant P_{i,t} < P_{i,\text{b}} \end{cases} \tag{36}$$

where $P_{i,max}$ is the rated output of the unit, $P_{i,\text{a}}$ is the lower limit of the output of its basic peak shaving, and generally takes 50% of the maximum power; $P_{i,\text{b}}$ and $P_{i,\text{c}}$ are the lower limits of the output of the first and second stages of DPS, respectively, which can take 40% and 30% of the maximum power.

#### 3.1.6. Compensation income of DPS

Due to the participation in DPS, the thermal power unit can be compensated by the power grid company according to the unit power compensation price $C_{\text{com}}$. The calculation is as follows:

$$G_{i,t,t\in T_{i,\text{dp}}}^{\text{com}} = \left(\left(P_{i,\text{a}} - P_{i,t}\right)\Delta T\right) C_{\text{com}} \tag{37}$$

where $T_{i,\text{dp}}$ represents the time period set of thermal power units for deep peak shaving.

In summary, the total operating costs considering the DPS of thermal power units are expressed as follows:

$$S_{i,t}^{\text{c}} = S_{i,t}^{\text{coal}} + S_{i,t}^{\text{T}} + S_{i,t}^{\text{add}} - G_{i,t,t\in T_{i,\text{dp}}}^{\text{com}} \tag{38}$$

In addition, in order to measure the total economy of thermal power units participating in DPS, the unit power generation costs $C_{\text{power}}^{\text{thermal}}$ are calculated as follows:

$$C_{\text{power}}^{\text{thermal}} = \left(\sum_{t=1}^{T}\sum_{i=1}^{N} S_{i,t}\right) \Big/ \left(\sum_{t=1}^{T}\left(P_{i,t}\Delta T\right)\right) \tag{39}$$

### 3.2. Collaborative optimization model

#### 3.2.1. Objective function

From a societal perspective, taking into account the comprehensive costs model for EAL and the whole process costs model for thermal power DPS, while also incorporating the penalty costs associated with RE curtailment, this research establishes the objective function as follows:

$$F = min \sum_{t=1}^{T} \left( \sum_{i=1}^{N} S_{i,t}^{\text{c}} + \sum_{j=1}^{J} S_{j,t}^{\text{al}} + \theta\left(P_{\text{w},t}^{\text{pr}} - P_{\text{w},t}\right) \right) + \alpha\left(P_{\text{v},t}^{\text{pr}} - P_{\text{v},t}\right) \tag{40}$$

where $N$ is the number of thermal power units, $J$ is the number of EAL,

and $\theta$ is the penalty coefficient of wind abandonment. $P_{w,t}^{pr}$ and $P_{w,t}$ are wind power prediction power and consumption power, respectively; $P_{v,t}^{pr}$ and $P_{v,t}$ are the photovoltaic (PV) predicted power and the consumed power, respectively.

### 3.2.2. Constraints

The constraint conditions include Eqs. (1)–(3), (8)–(21), and other constraints are as follows.

(1) Power balance constraints

$$\sum_{i=1}^{N} P_{i,t} + P_{hyd,t} + P_{w,t} + P_{line,t} = P_{load,t} + \sum_{j=1}^{J} P_{j,t}^{al} \tag{41}$$

where $P_{hyd,t}$ is the power of hydropower (Hyd) contained in the system. This paper studies the contribution of EAL regulation to RE consumption on the basis of the original system. Hyd does not participate in system optimization but is replaced by historical output data of hyd. $P_{line,t}$ is the tie-line power between the regional power grid and the large power grid.

(2) Wind power output constraints.

Wind power uses day-ahead forecast data, and its output range is:

$$0 \le P_{w,t} \le P_{w,t}^{pr} \tag{42}$$

(3)Thermal power operating characteristic constraints

$$\begin{cases} U_{i,t}P_{i,min} \le P_{i,t} \le U_{i,t}P_{i,max} \\ (U_{i,t} - U_{i,t-1})\left(T_{i,t-1}^{off} - T_{i,off}\right) \ge 0 \\ (U_{i,t-1} - U_{i,t})\left(T_{i,t-1}^{on} - T_{i,on}\right) \ge 0 \\ -V_i \le P_{i,t} - P_{i,t-1} \le V_i \end{cases} \tag{43}$$

where $P_{i,min}$ and $P_{i,max}$ are the upper and lower limits of thermal power unit output; $T_{i,off}$ and $T_{i,on}$ are the minimum continuous downtime and running time of the unit, and $V_i$ is the maximum climbing power of the unit.

(4) System spinning reserve constraints

$$\begin{cases} \sum_{i=1}^{N} U_{i,t}\left(P_{i,max} - P_{i,t}\right) \ge \alpha_{load}P_{load,t} + \alpha_w P_{w,t} + \beta_v P_{v,t} \\ \sum_{i=1}^{N} U_{i,t}\left(P_{i,t} - P_{i,min}\right) \ge \alpha_{load}P_{load,t} + \alpha_w P_{w,t} + \beta_v P_{v,t} \end{cases} \tag{44}$$

where $\alpha_{load}$, $\alpha_w$ and $\beta_v$ are the spinning reserve coefficients of system load forecasting, wind power forecasting and PV forecasting, respectively, usually taking 0.5,0.15 and 0.15.

## 3.3. Model solving

The current model established is a mixed-integer nonlinear programming (MINLP) model. To reduce the difficulty of solving, it is necessary to linearize the quadratic term in Eq. (5), as well as the piecewise functions in Eqs. (29) and (36). Here, this research adopts a piecewise linearization method. Taking the transformation of Eq. (29) as an example, the mathematical processing is shown below.

Four 0/1 variables $y_1 \sim y_4$ are set to represent the four segments of the EAL costs model. Suppose 5 continuous variables $w_1 \sim w_5$, then the research in this study have the following constraints:

$$\begin{cases} \sum_{i}^{I=4} y_{i,t} = 1 \\ \sum_{k}^{K=5} w_{k,t} = 1, \forall w_{k,t} \ge 0 \\ w_{1,t} \le y_{1,t} \\ w_{2,t} \le y_{1,t} + y_{2,t} \\ w_{3,t} \le y_{2,t} + y_{3,t} \\ w_{4,t} \le y_{3,t} + y_{4,t} \\ w_{5,t} \le y_{4,t} \\ P_{j,t}^{al} = w_1 P_{j,I_{j,min}^{al}}^{al} + w_2 P_{j,0.7I_{j,N}^{al}}^{al} + w_3 P_{j,0.9I_{j,N}^{al}}^{al} \\ \qquad + w_4 P_{j,I_{j,N}^{al}}^{al} + w_5\left(\omega_{al} P_{j,N}^{al}\right) \\ S_{j,t}^{al} = w_1 S_{j,I_{j,min}^{al}}^{al} + w_2 S_{j,0.7I_{j,N}^{al}}^{al} + w_3 S_{j,0.9I_{j,N}^{al}}^{al} \\ \qquad + w_4 S_{j,I_{j,N}^{al}}^{al} + w_5 S_{j,I_{j,max}^{al}}^{al} \end{cases} \tag{45}$$

The formula indicates that the load power at each moment can be weighted by five segmented points. The comprehensive costs of EAL at any time can be obtained by weighting the costs corresponding to the five segmentation points.

Additionally, for Eq. (19), this research adopts the classic Large Number Theorem [36] to handle it. Given 0/1 variables *slack*, relaxation variable $re = 0.0001$, and a large number $M_1$, the following constraints are introduced:

$$\begin{cases} 0.9I_{d,N,j}^{al} \ge I_{d,j,t}^{al} - M_1\left(1 - slack_{j,t}\right) + re \\ 0.9I_{d,N,j}^{al} \le I_{d,j,t}^{al} + M_1 slack_{j,t} \\ -M_1\left(1 - slack_{j,t}\right) \le M_{j,t}^{al} \le M_1\left(1 - slack_{j,t}\right) \end{cases} \tag{46}$$

The coordination optimization model in this paper is based on MATLAB R2023a, solved using GUROBI 11.0 solver, implemented in an environment with Intel Core i5-12500H and 16GB of RAM.

# 4. Case study

## 4.1. Case description

This research takes an actual regional power grid in Yunnan, China as an example to verify the effect of the proposed model. The regional power grid has two characteristics: 1) It is rich in hydropower resources and a high proportion of wind power and photovoltaic installed capacity, and the proportion of thermal power installed capacity is relatively low; 2) The proportion of electrolytic aluminum load in the total regional load is as high as 71.16%. Since the Hyd power is also a clean energy source, it faces the limitation of abandoned water and the limited ability to regulate and support the system. Obviously, the peak shaving resources in the regional power grid are scarce and need to be further explored.

The installed capacity and load scale of various types of power sources in the regional power grid are shown in Table 1. The relevant parameters of thermal power units are obtained from an actual regional power grid survey, as shown in Table 3. Post-flexibility transformation, many thermal power units no longer require oil injection for operation during PDS. However, a subset of older units and those yet to undergo transformation still rely on oil injection. To enhance the research

**Table 1**
Capacity of various types of power sources and the size of loads in an actual regional power grid.

| | Unit capacity /MW | Number of units /units | Capacity /MW |
|---|---|---|---|
| Thermal power unit(2530 MW) | 120 | 5 | 600 |
| | 130 | 1 | 130 |
| | 300I | 3 | 900 |
| | 300II | 3 | 900 |
| Hydroelectricity | – | – | 4622 |
| Wind power | – | – | 3500 |
| Photovoltaic power | – | – | 500 |
| Conventional loads | – | – | 2200 |
| EA Rated Load | – | – | 5428.8 |

framework, this research proposes integrating fuel consumption costs into the DPS analysis, albeit with adjustments based on actual conditions. The actual regional power grid survey reveals that only 120 MW units necessitate oil injection for DPS, while other units operate with zero fuel consumption. Parameters related to the EAL are listed in Table 2, where its maximum upward regulation power is set at 105% of its rated value. The time-of-use tariff data utilized comes from Reference [10], the spot price for aluminum is RMB 19,000 per ton. Due to the significant RE absorption pressure during the flood season, Hyd power, conventional load forecasting, and PV forecasting are based on a typical day's data from June of historical records. To comprehensively illustrate the impacts of various schemes, this study opts for three wind power forecasting scenarios for analysis. Among these, wind power scenario 1 exemplifies a robust anti-peaking characteristic, wind power scenario 2 portrays a less pronounced anti-peaking feature, while wind power scenario 3 displays a flat peaking characteristic. The Hyd output, conventional load, wind power forecast, and PV forecast scenarios for these tests can be found in Fig. 3. Two comparison schemes are set in each scenario:

Case1: The EAL is not involved in the regulation.

Case2: Collaborative optimization of EAL regulation and thermal power DPS.

Additionally, due to the increase in capacity prices and transformer capacity constraints that result from raising the power of EAL, the willingness and ability of aluminum smelters to adjust their power consumption upwards may vary. To further explore the impact of the adjustment space of EAL on its comprehensive cost and the total social peak shaving benefit, this paper tests the effect of the up-regulation space between 101% and 110% of the rated power based on the Case2 model. The main work of this paper is to construct a refined regulation

**Table 2**
Parameters related to EAL.

| Aluminum Plant Serial No. | Series No. | Rated Power /MW | Rated Current /kA | Reverse electromotive force/v | Slot resistance /mΩ |
|---|---|---|---|---|---|
| 1 | EA1 | 390.4 | 400 | 472 | 1.26 |
| 2 | EA2 | 494.4 | 400 | 708 | 1.32 |
| 3 | EA3 | 496 | 400 | 720 | 1.30 |
| | EA4 | 496 | 400 | 720 | 1.30 |
| | EA5 | 496 | 400 | 720 | 1.30 |
| 4 | EA6 | 520 | 400 | 496 | 2.01 |
| | EA7 | 520 | 400 | 496 | 2.01 |
| | EA8 | 520 | 400 | 496 | 2.01 |
| 5 | EA9 | 744 | 500 | 858 | 1.26 |
| | EA10 | 752 | 500 | 814 | 1.38 |

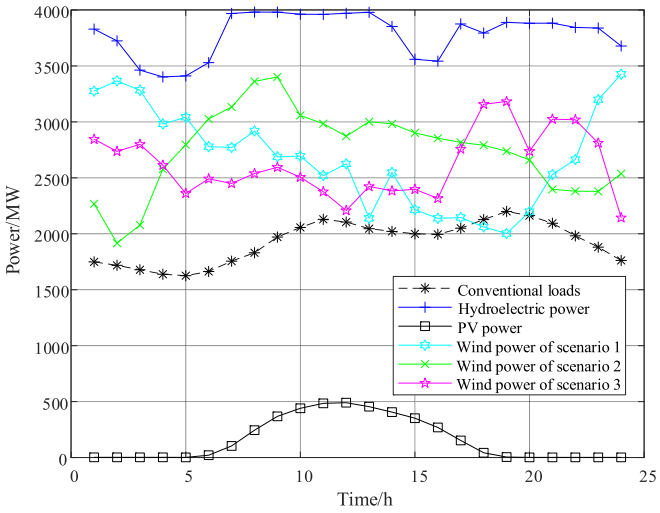



**Fig. 3.** Installed capacity of various types of power sources and load size of regional power grids.

model of EAL and study its collaborative optimization strategy with thermal power DPS. The market price is considered a deterministic factor, represented by the fixed peak-valley price and the spot price of aluminum. However, in fact, both the time-of-use electricity price and aluminum price are volatile and uncertain, impacting the comprehensive cost of EAL and even the overall social peak shaving economy. Future research will further consider market price fluctuations in detail.

### 4.2. Results analysis

#### 4.2.1. Analysis of social total peak shaving effect

Under the three wind power scenarios, Table 4 presents comprehensive indicators for the peak regulation performance of Case1 and Case2. Regarding RE accommodation, the PV abandonment rate for both schemes is zero in all three scenarios, though not explicitly listed in the table. However, Case2 consistently exhibits superior wind power abandonment (WPA) rates compared to Case1, with WPA rates decreasing by 4.61%, 4.07%, and 3.16% respectively. On average, there is a relative decrease of 60%. For Case1, without considering the adjustment of EAL, ensuring the consumption of RE solely through the DPS capacity of thermal power units is less effective. Even with substantial utilization of the DPS capabilities of thermal power plants, there remains a significant amount of WPA, and this also leads to increased costs of thermal power DPS. In contrast, Case2 optimally coordinates EAL regulation with thermal power DPS, alleviating pressure on thermal power plants. Compared to Case1 under the three wind power scenarios, the costs of thermal power DPS in Case2 are reduced by 12.53%, 14.06%, and 22.10% respectively. Furthermore, under all three scenarios, coal consumption costs associated with thermal power in Case2 surpass those in Case1. This increase is attributed to the EAL regulation to better accommodate RE fluctuations in Case2, resulting in greater electricity generation and higher coal consumption of thermal power for grid balance.

Regarding the system peak shaving economy, both the total social peak shaving cost and the total operating cost of thermal power are significantly reduced. Compared with Case1, the total social peak shaving cost in Case2 under three wind power scenarios decreases by 9.69%, 7.16%, and 5.45%, respectively, while the total operating cost of thermal power is reduced by 7.52%, 5.36%, and 8.52%, respectively. However, the comprehensive cost of EAL in Case2 across the three wind power scenarios increases by 9.24%, 10.13%, and 10.00%, respectively. From a specific economic evaluation perspective, in wind power scenario 1, the comprehensive cost of EAL increases by $924,884.34, while the total social peak shaving cost is reduced by $1,581,258.18,

**Table 3**
Relevant parameters of thermal power units.

| Unit capacity /MW | Rate of climb /(MW·h) | Minimum start/stop time/(h) | Consumption characteristic costs factor a/[$/(MW·h)$^2$] | Consumption characteristic costs factor b/($/MW) | Consumption characteristic costs factor c/$ | Start-up and shutdown costs/$ | Deep Peak Fuel Throw /(t/h) |
|---|---|---|---|---|---|---|---|
| 120 | 45 | 5 | 0.048 | 23.23 | 639.4 | 1000 | 0.55 |
| 130 | 50 | 5 | 0.063 | 16.51 | 502.7 | 1100 | 0 |
| 300I | 80 | 8 | 0.079 | 22.10 | 460.0 | 1500 | 0 |
| 300II | 80 | 8 | 0.070 | 23.90 | 471.0 | 2000 | 0 |

**Table 4**
Comprehensive indicators for system peaking.

| Wind power scenarios | Case | Coal costs of thermal power /$ | costs of thermal power DPS /$ | Total operating costs of thermal power /$ | Electricity costs of EAL /$ | Incremental costs of EAL Capacity Tariffs/$ | Comprehensive costs of EAL /$ | Total social peaking costs /$ | WPA rate |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Case1 | 902,842.81 | 908,405.06 | 1,811,247.87 | 10,010,283.75 | 0.00 | 10,010,283.75 | 16,319,451.68 | 8.7557% |
| | Case2 | 880,469.24 | 794,557.79 | 1,675,027.03 | 10,145,829.84 | 1,258,531.56 | 10,935,168.10 | 14,738,193.50 | 4.1424% |
| 2 | Case1 | 873,458.56 | 864,951.79 | 1,738,410.35 | 10,010,283.75 | 0.00 | 10,010,283.75 | 17,104,314.68 | 10.1555% |
| | Case2 | 901,815.05 | 743,371.79 | 1,645,186.84 | 10,284,498.26 | 1,258,531.56 | 11,024,105.53 | 15,879,826.73 | 6.0879% |
| 3 | Case1 | 831,685.07 | 957,331.28 | 1,789,016.35 | 10,010,283.75 | 0.00 | 10,010,283.75 | 13,623,122.16 | 3.6257% |
| | Case2 | 890,959.75 | 745,721.84 | 1,636,681.59 | 10,335,951.56 | 1,258,531.56 | 11,011,356.30 | 12,880,032.97 | 0.4612% |

**Table 5**
Economic and Technical Indicators of Deep Peaking of Thermal Power Units.

| Wind power scenarios | Case | Wear costs /$ | Oil consumption costs /$ | Environmental penalty costs /$ | Power loss opportunity costs /$ | Compensation income /$ | unit generation costs /($/kW·h) | Deep peaking I stage /times | Deep peaking II stage /times |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Case1 | 632,810.51 | 49,534.61 | 80,848.08 | 442,777.11 | 297,565.26 | 0.0778 | 6 | 204 |
| | Case2 | 566,985.07 | 30,436.93 | 64,960.91 | 403,025.05 | 270,850.16 | 0.0729 | 8 | 168 |
| 2 | Case1 | 616,513.87 | 36,404.96 | 70,521.42 | 431,494.21 | 289,982.67 | 0.0768 | 14 | 181 |
| | Case2 | 528,977.55 | 27,452.92 | 60,989.11 | 384,051.00 | 258,098.79 | 0.0695 | 4 | 158 |
| 3 | Case1 | 669,558.32 | 45,953.80 | 83,495.95 | 482,756.05 | 324,432.83 | 0.0841 | 0 | 213 |
| | Case2 | 539,614.37 | 22,678.50 | 57,899.94 | 382,760.67 | 257,231.64 | 0.0701 | 20 | 151 |

**Table 6**
Effect of different up-regulation spaces of EAL in Case2.

| Wind power scenarios | upward space | Comprehensive costs of EAL/$ | Total societal peaking costs /$ | WPA rate |
|---|---|---|---|---|
| 1 | 102% | 10,325,372.97 | 15,492,754.52 | 6.6060% |
| | 104% | 10,730,969.91 | 14,969,295.96 | 4.9233% |
| | 106% | 11,147,620.79 | 14,571,856.86 | 3.4037% |
| | 108% | 11,599,613.65 | 14,422,198.59 | 2.2931% |
| | 110% | 12,002,315.41 | 14,517,949.34 | 1.6905% |
| 2 | 102% | 10,360,653.90 | 16,526,843.71 | 8.3403% |
| | 104% | 10,781,762.28 | 16,047,738.34 | 6.7782% |
| | 106% | 11,262,569.87 | 15,764,597.91 | 5.4183% |
| | 108% | 11,685,593.73 | 15,483,270.92 | 4.1006% |
| | 110% | 12,186,495.48 | 15,356,499.84 | 2.7830% |
| 3 | 102% | 10,390,488.49 | 12,956,547.43 | 1.6648% |
| | 104% | 10,803,053.30 | 12,741,249.58 | 0.6238% |
| | 106% | 11,199,713.88 | 13,052,880.13 | 0.4098% |
| | 108% | 11,356,935.77 | 13,215,115.28 | 0.4098% |
| | 110% | 11,449,260.61 | 13,285,715.06 | 0.4099% |

representing the total social peak shaving benefit generated. Similarly, in wind power scenario 2, the total social peak shaving benefit generated is $1,224,487.95, while the EAL cost increase is $1,013,821.77. The corresponding benefit and cost values in wind power scenario 3 are $666,574.73 and $380,204.73, respectively. Therefore, the total social peak shaving benefits provided by the proposed scheme in the three wind power scenarios substantially exceed the incremental comprehensive cost of EAL, indicating good economic efficiency. In summary, the participation of EAL in regulation can effectively reduce the WPA of the system, and effectively reduce the total operating cost and deep shaving cost of thermal power. Although the comprehensive cost of EAL has increased, the total social peak regulation benefit obtained is sufficient to cover, indicating that the proposed scheme provides effective peak regulation and economic efficiency.

#### 4.2.2. Analysis of thermal power DPS

The economic and technical indicators for the profound peak shaving of thermal power units under various scenarios are showed in Table 5. Compared to Case1, Case2 displays a notable reduction in the supplementary costs accrued during thermal power.

DPS periods. This results in a substantial enhancement in the economic performance of thermal power units. Under the three wind power scenarios, the unit generation costs for thermal power decrease by 6.31%, 9.45% and 16.66%, respectively. However, within the

comprehensive cost framework of thermal power DPS, the compensation income for DPS proves insufficient to cover the escalated costs. During the scheduling period, the frequency and depth of DPS for thermal power units decreases in Fig. 4(c), Fig. 5(c) and Fig. 6(c). The figures unequivocally reveal that, across all scenarios, the thermal power units in Case2 undergo a significantly lower total number of DPS events and exhibit shallower shaving depths compared to Case1. In the context of two-stage DPS, Case2 records reductions of 32, 33, and 42 instances of the deep peak shaving, respectively. However, it's crucial to note that employing DPS can elevate both coal consumption and carbon emissions, underscoring the need to limit its use when feasible. Additionally, enhancing the production technology of thermal power units remains essential to mitigate.

these effects. To summarize, integrating EAL in grid regulation markedly enhances the economic efficiency of thermal power DPS, and diminishes its frequency and depth, consequently reducing the unit generation cost of thermal power.

#### 4.2.3. Analysis of the regulation of EAL

The power regulation diagrams for EAL under various wind scenarios are showed in Fig. 4(b), Fig. 5(b) and Fig. 6(b). Upon EAL engagement in regulation, a notable increase in capacity price costs is observed, remaining consistent across all three wind scenarios. This indicates that EAL have been adjusted to their maximum power output levels in each case, resulting in a significant cost escalation. Compared with Case1, the costs of EAL in the three wind power scenarios have risen by 9.24%, 10.13% and 10.00%, respectively. Fig. 7 and Fig. 8 illustrate the temperature and current changes of EAL in the three.

scenarios. Fig. 7 demonstrates that EAL temperatures fluctuate within a narrow range between 956.5 °C and 962.5 °C under the three wind power scenarios, adhering to stringent production safety constraints.

Fig. 8 reveals the minimum current adjustment for EAL at 92% of its rated value, occurring under wind scenario 3, while the maximum current regulation reaches up to 103.3% of the rated current. The overall amplitude of current adjustments is relatively small, ensuring a safe and reliable operation for the EA production process. This modulation is practical and feasible to implement, showcasing the considerable potential for EAL to contribute to grid balancing by improving the overall social peak shaving effect. The cost increase associated with EAL primarily stems from the rise in capacity price costs under the current two-part tariff, while the increase in energy costs is comparatively minor. In order to encourage EA plants to participate in load regulation, providing electricity price subsidies is essential.

Although the participation of EAL in regulation has many advantages, in practice, improper regulation can lead to excessive up-regulation, resulting in overload and tripping, which disturb grid stability. Aluminum plants may face significant power outage losses and potential chain reactions affecting other loads. However, the proposed model in this paper carefully considers the safety of EAL regulation, which is a strong constraint, and allows independent adjustment boundaries to prevent instability and minimize risk. Nonetheless, there is still a possibility of risk, so detailed consideration should be given to setting adjustment boundaries in practical applications. Additionally, using EAL for peak regulation reduces its immediate support for system frequency.

#### 4.2.4. Analysis of different up-regulation space of EAL

Table 6 presents the impact of EAL in Case 2 under various up-regulation spaces, providing a detailed quantitative comparison. Evidently, as the up-regulation space increases, the WPA rate across the three wind power scenarios consistently decreases, while the comprehensive cost of EAL rises. Within the tested adjustment range, the trend of the total social peaking cost varies across the three wind power scenarios. Specifically, the total social peaking cost for scenarios 1 and 3 initially decreases and then increases, whereas for scenario 2, it continuously decreases. As previously mentioned, based on Case 1, the reduction in total social peak shaving cost due to the participation of EAL in regulation is considered the social total peak shaving benefit. There should be a cost-benefit balance point where the increased comprehensive cost of EAL is offset by the total social peak shaving benefit, demonstrating practical feasibility. Assuming that the

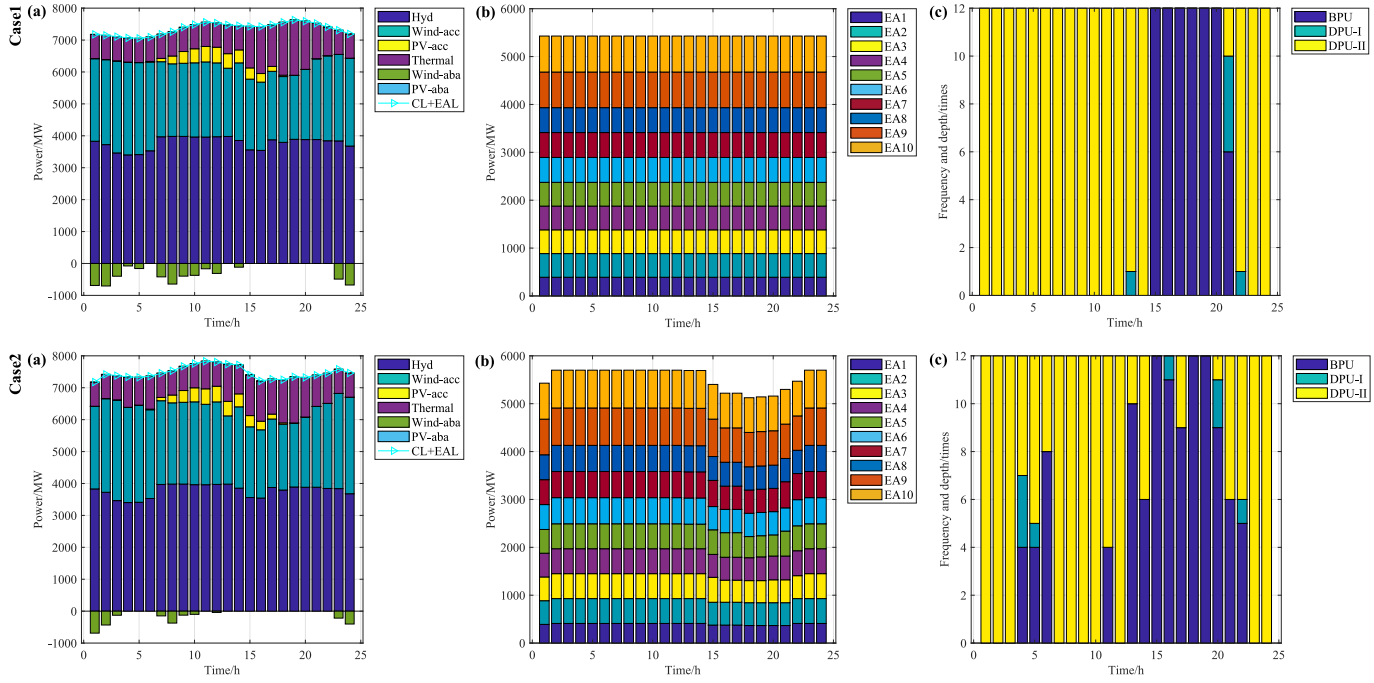



**Fig. 4.** Simulation results for wind power scenario 1: (a) power curves of each component; (b) power curves of each EAL; (c) frequency state of two-stage DPS of thermal power units;
Note: HP: Hyd power; WP-acc: Wind power accommodation; PV-acc: PV accommodation; WP-aba: Wind power abandonment;
PV-aba: PV abandonment. BPU: Basic peak-shaving unit; DPU-I: Units in the first stage of DPS; DPU-II: Units in the second stage of DPS.

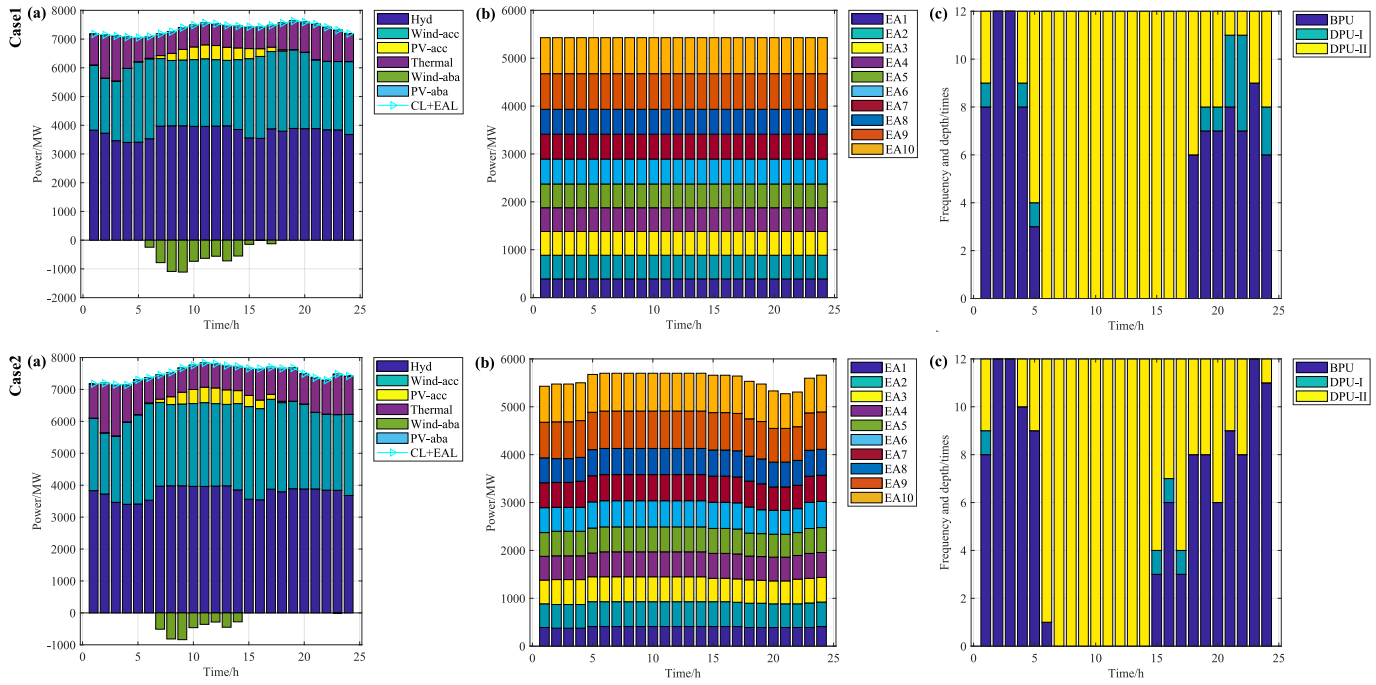



**Fig. 5.** Simulation results for wind power scenario 2: (a) power curves of each component; (b) power curves of each aluminum electrolysis load; (c) frequency state of two-stage deep peaking of thermal power units.

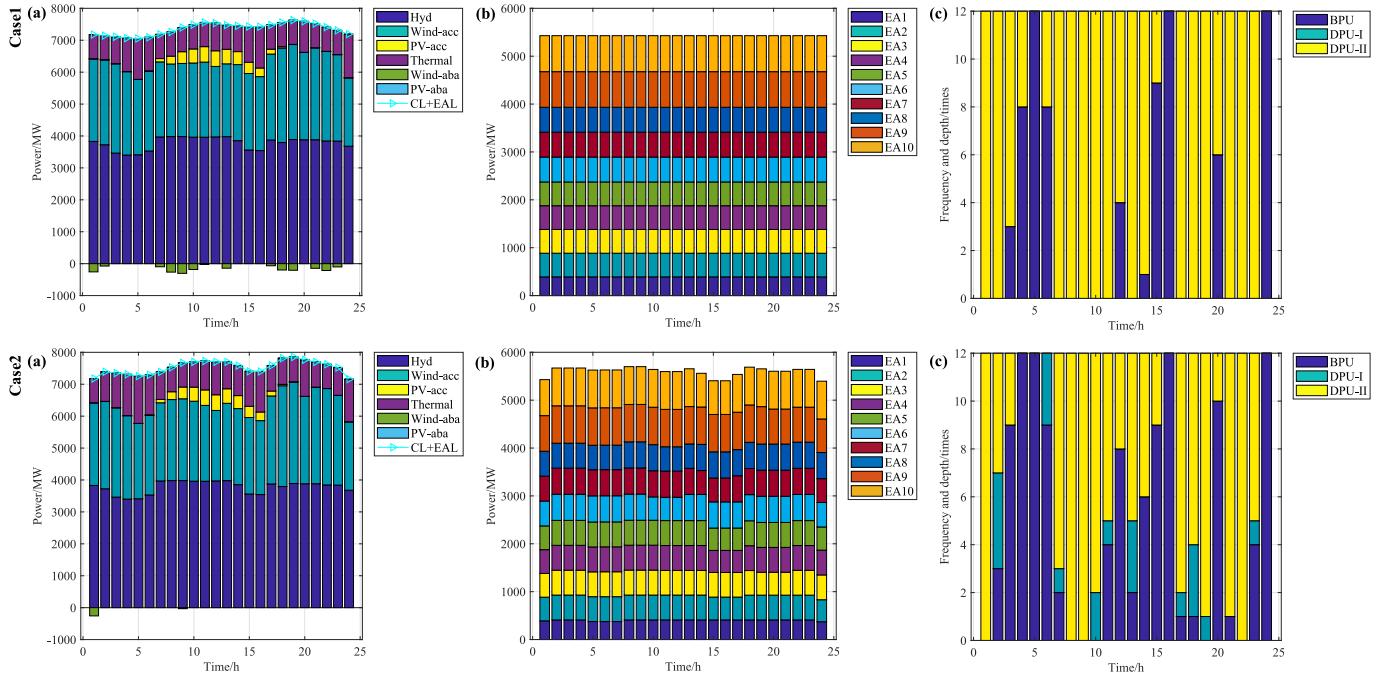



**Fig. 6.** Simulation results for wind power scenario 3: (a) power curves of each component; (b) power curves of each aluminum electrolysis load; (c) frequency state of two-stage deep peaking of thermal power units.

compensation ratio for EAL can be derived from the total social peak shaving benefit, different compensation ratios correspond to distinct cost-benefit balance points, defined by a specific up-regulation space value.

Fig. 9 shows the change of cost-benefit and WPA rate of different up-regulation Spaces in Case2, where the solid blue line is the comprehensive cost of EAL, and the dashed blue line is the compensation benefit under different ratios. Here, the cost-benefit balance points when the compensation ratio is 1.0, 0.9, 0.8, 0.7, and 0.6 are tested respectively. Obviously, the lower the compensation ratio, the smaller the up-regulation space values corresponding to the balance point. For instance, with a compensation ratio of 0.8, the up-regulation space values at the cost-benefit balance point in the three wind power scenarios are approximately 107.7%, 104.8%, and 103.5%, respectively. The position of the balance point is closely linked to the characteristics of each wind power scenario. Scenario 3 exhibits the weakest anti-peak regulation characteristic, resulting in the lowest equilibrium point, whereas Scenario 1 has the strongest anti-peak regulation characteristic

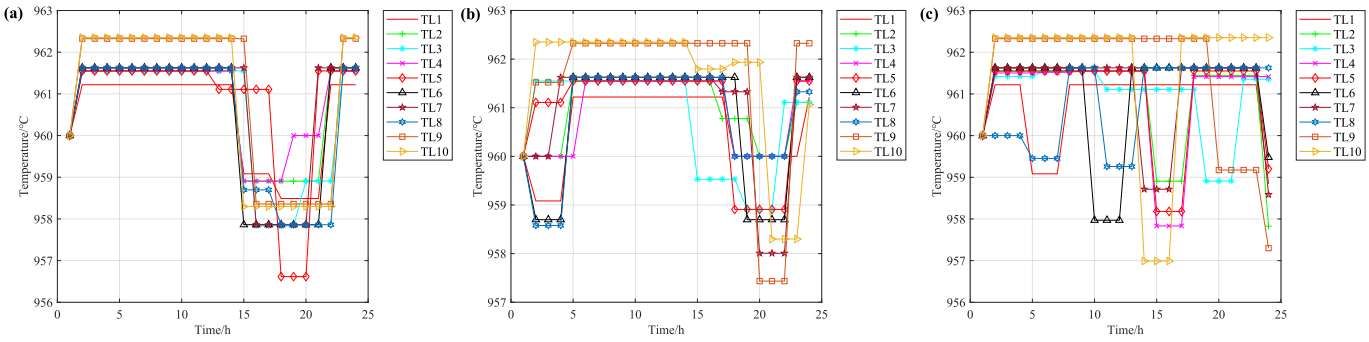



**Fig. 7.** Temperature variation curves for each EAL in Case2; (a)wind power scenario 1; (b)wind power scenario 2; (c)wind power scenario 3.

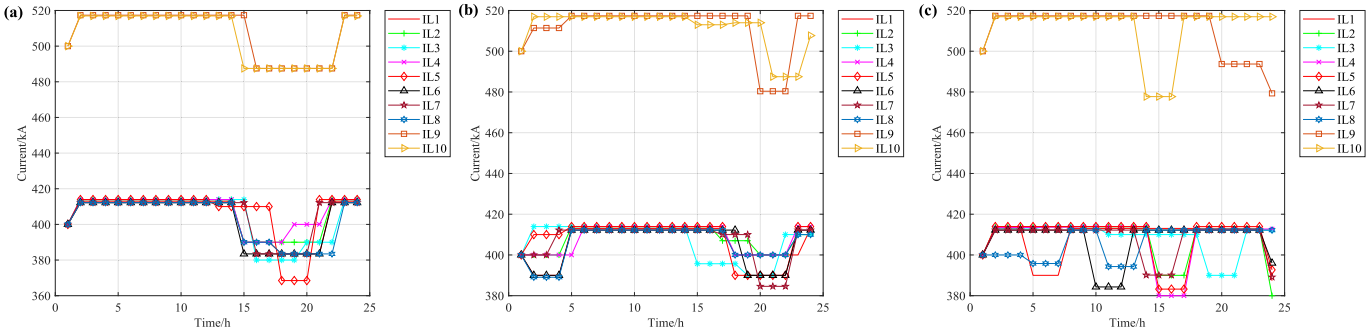



**Fig. 8.** Current variation curves for each EAL in Case2: (a) wind power scenario 1; (b) wind power scenario 2; (c) wind power scenario 3.

and the highest equilibrium point. Therefore, in practical applications, it is essential to consider the characteristics of wind power output and the cost-benefit balance point to appropriately set the up-regulation space of EAL, thereby ensuring its benefits and promoting its participation in regulation.

## 5. Conclusions

This research focuses on EAL as a typical energy-intensive load, developing a fine-tuned adjustment model considering production requirements and safety. The segmented cost characteristics of EAL are studied, and the cost of deep peaking of thermal power units is analyzed in detail. This study proposes a collaborative optimization method integrating EAL regulation and thermal power DPS for RE systems. Utilizing an actual regional power grid in China as a case study, this research validated the social peak regulation impact of the proposed model across three typical wind power scenarios. The main conclusions are as follows:

(1) Integrating the EAL regulation model substantially reduced RE curtailment rates by an average of 60.00% across scenarios. This led to significant decreases in overall social peak shaving costs, specifically by 9.69%, 7.16%, and 5.45%, enhancing the economic efficiency of thermal power DPS and resulting in a considerable decline in unit generation costs for thermal power.

(2) The proposed model successfully integrates the modulation capabilities of EAL with thermal power DPS, adeptly managing periods of energy surplus and shortage. This integration diminishes peak shaving pressure on thermal units, evident in the reduced frequency and depth of DPS events. However, the costs associated with EAL experienced a notable surge, averaging an increase of 9.79% across the three wind scenarios. This increase is primarily attributed to rising capacity price costs under the two-part tariff system.

(3) The quantitative results of the first two conclusions are based on the maximum up-regulation power of the EAL being 105% of the rated value. In the proposed scheme, the adjustment range of the EAL is relatively small, ensuring production safety and ease of implementation. Moreover, as the up-regulation space increases, the WPA rate decreases further. However, the comprehensive cost of the EAL rises, necessitating a cost-benefit balance between this cost and the total social peak shaving benefit. It is essential to allocate and compensate the EAL from the total social peak shaving benefit. The lower the distribution ratio, the lower the cost-benefit balance point and the smaller the corresponding up-regulation space. The cost-benefit balance point is closely related to the characteristics of the wind power scenario. These can guide the practical application of policy formulation.

(4) Given the limitations of current research and the potential for future expansion, future research work will focus more on the main benefits of EAL, comprehensively consider its participation in the power auxiliary service market, expand the scope of EAL regulation, and design hierarchical optimization strategies to improve system peaking efficiency. Additionally, considering the uncertainty of market prices, electricity price forecasting models and stochastic optimization methods can be developed to accurately predict electricity prices and optimize adjustment strategies, which is also one of the tasks worth carrying out in the future.

## CRediT authorship contribution statement

**Xiaoyu Yue:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis. **Siyang Liao:** Methodology, Data curation, Conceptualization. **Jian Xu:** Investigation, Funding acquisition, Data curation. **Deping Ke:** Supervision, Resources, Project administration. **Huiji Wang:** Visualization, Validation. **Jiaquan Yang:** Validation, Supervision. **Xuehao He:** Visualization, Validation, Supervision.

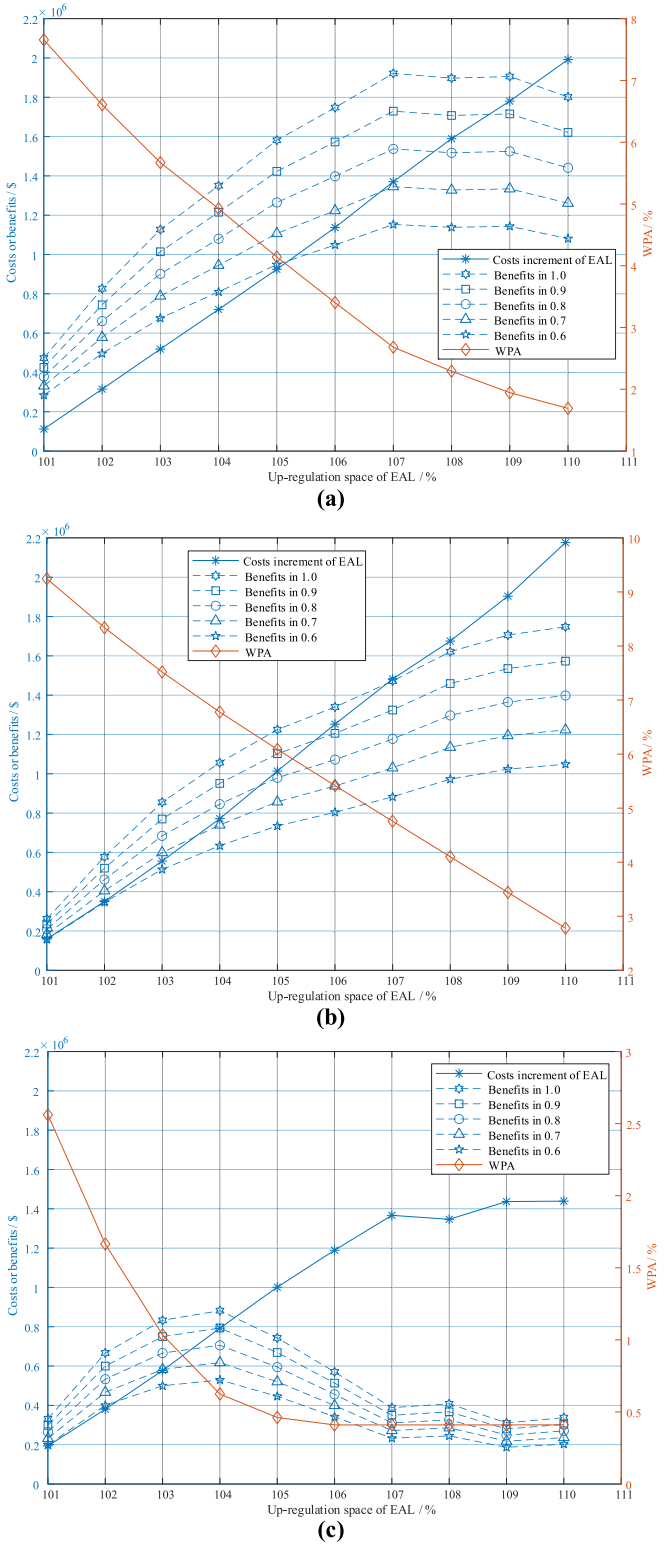



**Fig. 9.** Cost-benefit and WPA rate under different up-regulation spaces in Case2: (a) wind power scenario 1; (b) wind power scenario 2; (c) wind power scenario 3.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The data that has been used is confidential.

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## References

[1] Wang Y, Liu Y, Xu Z, et al. A review on renewable energy-based chemical engineering design and optimization[J]. Renew Sust Energ Rev 2024;189:114015.
[2] Li C, Chen D, Liu X, et al. Fault mitigation mechanism to pave the way to accommodate over 90% renewable energy in electric power systems[J]. Appl Energy 2024;359:122623.
[3] Luo H, Li Y, Gao X, et al. Carbon emission prediction model of prefecture-level administrative region: a land-use-based case study of Xi'an city, China[J]. Appl Energy 2023;348:121488.
[4] Yang T, Dong Q, Du Q, et al. Carbon dioxide emissions and Chinese OFDI: from the perspective of carbon neutrality targets and environmental management of home country[J]. J Environ Manag 2021;295:113120.
[5] Tarroja B, Mueller F, Eichman JD, et al. Spatial and temporal analysis of electric wind generation intermittency and dynamics[J]. Renew Energy 2011;36(12): 3424–32.
[6] Ueckerdt F, Brecha R, Luderer G. Analyzing major challenges of wind and solar variability in power systems[J]. Renew Energy 2015;81:1–10.
[7] Wei H, Hongxuan Z, Yu D, et al. Short-term optimal operation of hydro-wind-solar hybrid system with improved generative adversarial networks[J]. Appl Energy 2019;250:389–403.
[8] Li F, Chen S, Ju C, et al. Research on short-term joint optimization scheduling strategy for hydro-wind-solar hybrid systems considering uncertainty in renewable energy generation[J]. Energ Strat Rev 2023;50:101242.
[9] Xu J, Jiang X, Liao S, et al. Probabilistic prognosis of wind turbine faults with feature selection and confidence calibration[J]. IEEE Transactions on Sustainable Energy 2024;15(1):52–67.
[10] Lin L, Yue X, Xu B, et al. Sequence and strategy of pumped storage-thermal combined peak shaving considering benefits of pumped storage and deep regulation of thermal power[J]. Power Sys Techn 2021;45(01):20–32.
[11] Wang J, Zhang S, Huo J, et al. Dispatch optimization of thermal power unit flexibility transformation under the deep peak shaving demand based on invasive weed optimization[J]. J Clean Prod 2021;315:128047.
[12] Lin L, Xu B, Xia S. Multi-angle economic analysis of coal-fired units with plasma ignition and oil injection during deep peak shaving in China[J]. Appl Sci 2019;9 (24):5399.
[13] Gan L, Yang T, Chen X, et al. Purchased power dispatching potential evaluation of steel plant with joint multienergy system and production process optimization[J]. IEEE Trans Ind Appl 2022;58(2):1581–91.
[14] Liu K, Guan X, Gao F, et al. Self-balancing robust scheduling with flexible batch loads for energy intensive corporate microgrid[J]. Appl Energy 2015;159:391–400.
[15] Pan R, Li Z, Cao J, et al. Electrical load tracking scheduling of steel plants under time-of-use tariffs[J]. Comput Ind Eng 2019;137:106049.
[16] Cao J, Pan R, Xia X, et al. An efficient scheduling approach for an iron-steel plant equipped with self-generation equipment under time-of-use electricity tariffs[J]. Swarm and Evolutionary Comput 2021;60:100764.
[17] Wang J, Wang Q, Sun W. Optimal power system flexibility-based scheduling in iron and steel production: a case of steelmaking–refining–continuous casting process [J]. J Clean Prod 2023;414:137619.
[18] Su P, Zhou Y, Wu J. Multi-objective scheduling of a steelmaking plant integrated with renewable energy sources and energy storage systems: balancing costs, emissions and make-span[J]. J Clean Prod 2023;428:139350.
[19] Chen R, Sun H, Guo Q, et al. Profit-seeking energy-intensive enterprises participating in power system scheduling: model and mechanism[J]. Appl Energy 2015;158:263–74.
[20] Jin H, Li Z, Sun H, et al. A robust aggregate model and the two-stage solution method to incorporate energy intensive enterprises in power system unit commitment[J]. Appl Energy 2017;206:1364–78.
[21] Nosratabadi SM, Hooshmand RA, Gholipour E. Stochastic profit-based scheduling of industrial virtual power plant using the best demand response strategy[J]. Appl Energy 2016;164:590–606.
[22] Ramin D, Spinelli S, Brusaferri A. Demand-side management via optimal production scheduling in power-intensive industries: the case of metal casting process[J]. Appl Energy 2018;225:622–36.
[23] Xu J, Liao S, Sun Y, et al. An isolated industrial power system driven by wind-coal power for aluminum productions: a case study of frequency control[J]. IEEE Trans Power Syst 2015;30(1):471–83.
[24] Liao S, Xu J, Sun Y, et al. Load-damping characteristic control method in an isolated power system with industrial voltage-sensitive load[J]. IEEE Trans Power Syst 2016;31(2):1118–28.
[25] Xu J, Chen Y, Liao S, et al. Demand side industrial load control for local utilization of wind power in isolated grids[J]. Appl Energy 2019;243:47–56.

[26] Aguero JL, Beroqui M, Achilles S. Aluminum plant. Load modeling for stability studies[C]//1999 IEEE Power Engineering Society Summer Meeting. Conference Proceedings (Cat. No.99CH36364): Vol. 2. 1999. 1330–1335 vol.2.
[27] Liao S, Xu J, Sun Y, et al. Control of energy-intensive load for power smoothing in wind power plants[J]. IEEE Trans Power Syst 2018;33(6):6142–54.
[28] Liao S, Xu J, Sun Y, et al. Local utilization of wind electricity in isolated power systems by employing coordinated control scheme of industrial energy-intensive load[J]. Appl Energy 2018;217:14–24.
[29] Cai G, Zhou J, Wang Y, et al. Multi-objective coordinative scheduling of system with wind power considering the regulating characteristics of energy-intensive load[J]. Int J Electr Power Energy Syst 2023;151:109143.
[30] Yi Bao. Study on the strategy of high energy-consuming electrolytic aluminum load participating in power system frequency modulation and auxiliary service [D / OL].Wuhan University. 2019 [2023−10−13].
[31] Jia Zhang. Thermal field simulation of 500 kA electrolytic cell in an electrolytic aluminum plant [J]. Nonferrous metallurgical design and research 2018;39(3): 51–4.
[32] Zhuxian Qiu. Prebaked Aluminium Smelting (Third Edition) [M]. Beijing: Metallurgical Industry Press; 2005.
[33] Morishige T, Haarberg GM, Gudbrandsen H, et al. Effects of composition and temperature on current efficiency for Aluminium electrolysis from Cryolite-based molten alumina electrolytes[J]. ECS Trans 2017;77(11):997.
[34] Čepin M. Optimization of safety equipment outages improves safety[J]. Reliab Eng Syst Saf 2002;77(1):71–80.
[35] Song JS, Lok V, Yoon KB, et al. Quantitative risk-based inspection approach for high-energy piping using a probability distribution function and modification factor[J]. Int J Press Vessel Pip 2021;189:104281.
[36] Lesage-Landry A, Wang H, Shames I, et al. Online convex optimization of multi-energy building-to-grid ancillary services[J]. IEEE Trans Control Syst Technol 2020;28(6):2416–31.